Some basic definitions on components of a computer system

Input and Output Devices (also known as I/O devices)

Input Devices are the devices that are used to send signals to the computer for performing tasks.

Examples are keyboard, mouse, scanner, etc.

Output Devices are the devices that show us the result generated from a computer system. Output can be of many different forms like image, audio, video, etc.

Examples are monitor, printer, speaker, etc.

Operating system

An **operating system** is a system software that manages computer hardware and software resources. It also provides a basis for application programs and acts as an intermediary between the computer user and the computer hardware.

Some examples of operating systems are Microsoft Windows, Google's Android OS, Linux Operating System, etc.

Assembler

An assembler is a type of computer program that converts assembly level language code into machine language code.

Compiler

A compiler is a type of computer program that converts a program written in a high-level language to a low-level language.

Interpreter

An interpreter is a type of computer program that directly executes instructions written in a programming language, without needing to compile the program first into a machine language program. An interpreter translates code written in a high-level programming language into machine code line-by-line as the code runs.

Linker

A linker is a computer system program that takes the object code generated by the compiler/assembler and combines it with other necessary libraries and modules to create an executable file. It forms an executable file as output for the loader.

Loader

A loader is a software component in an operating system that loads executable programs into memory and prepares them for execution by the CPU. Loader takes input of executable files from linker, loads it to main memory, and prepares this code for execution by computer. Loader allocates memory space to program.

Differences between Compiler and Interpreter

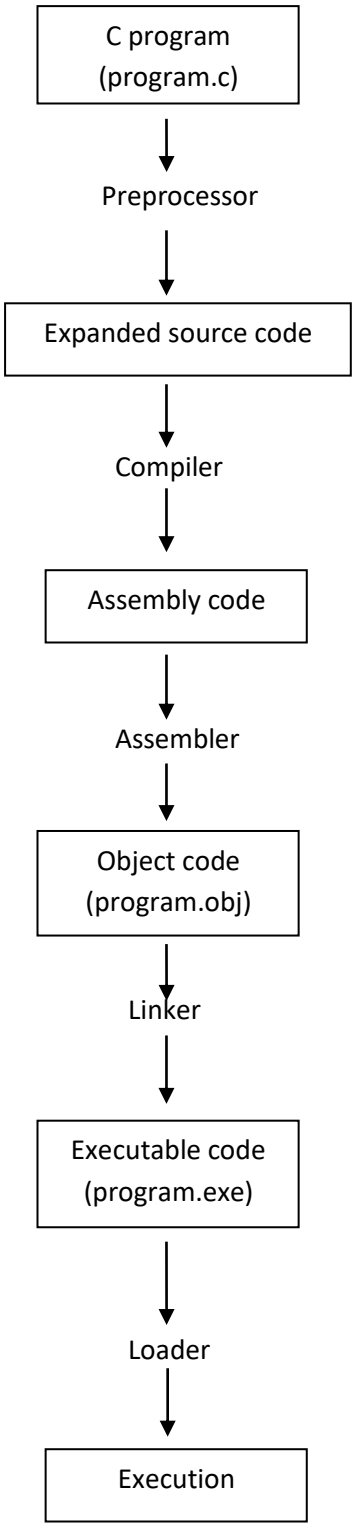
A Compiler translates the entire source code into machine code before execution; whereas an Interpreter translates and executes the source code line-by-line or statement-by-statement.

Compiled code runs faster, while interpreted code runs slower.

In case of compiler, errors are detected and must be corrected after the entire program is compiled. In case of interpreter, errors are detected and must be corrected line-by-line during execution.

Examples of Programming languages that use compilers are C, C++, etc. Examples of Programming languages that use interpreter are Python, Ruby, etc.

What are the different steps in the execution of a C Program?



Explanation of the above steps

At first, the source code (C program) is sent to preprocessor. The preprocessor generates an expanded source code. The expanded source code is sent to compiler which compiles the code and converts it to

assembly code. The assembly code is then sent to assembler which converts it into object code. The object code is then sent to linker which links it to the libraries. Then it is converted into executable code. The executable code is then sent to loader which loads it into computer memory and then it is executed.